

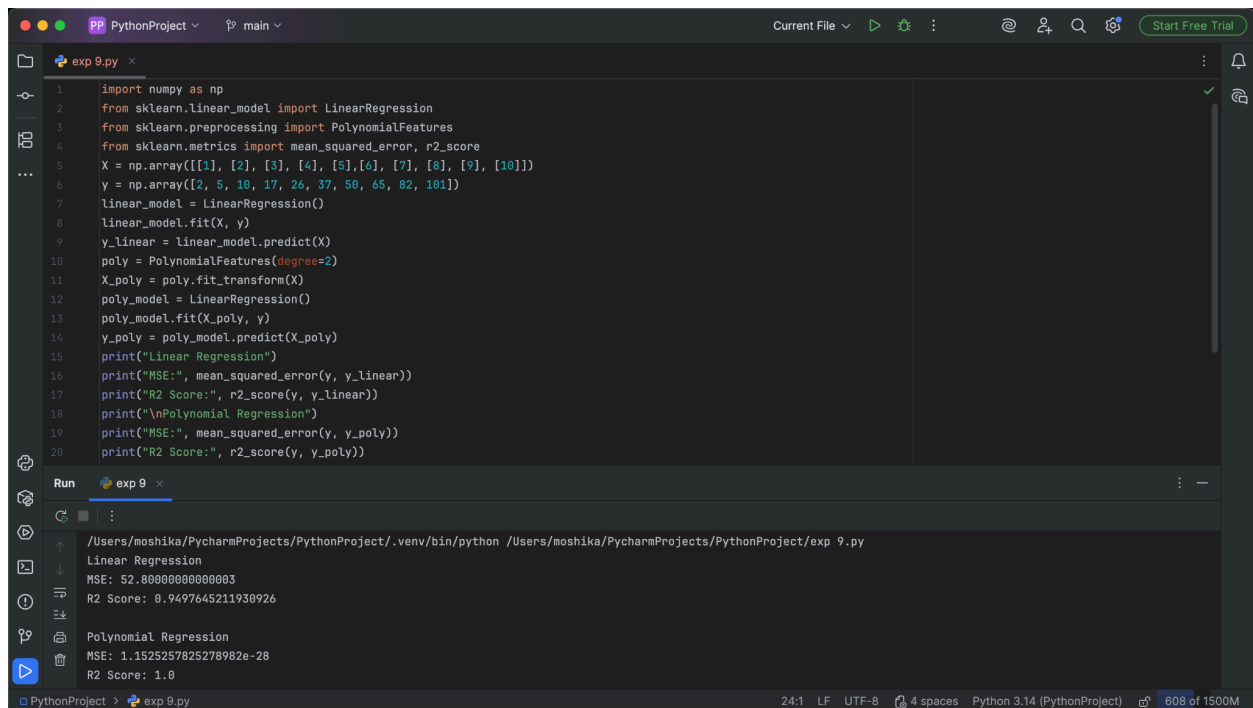
EXPERIMENT - 9

Aim: To compare Linear Regression and Polynomial Regression using Python.

Procedure:

1. Import the required libraries.
2. Load the dataset.
3. Select the input and target variables.
4. Split the dataset into training and testing sets.
5. Create and train the Linear Regression model.
6. Transform the input features into polynomial features.
7. Create and train the Polynomial Regression model.
8. Generate predictions using both models.
9. Calculate and compare their performance.
10. Display the comparison results.

Output:



```
1 import numpy as np
2 from sklearn.linear_model import LinearRegression
3 from sklearn.preprocessing import PolynomialFeatures
4 from sklearn.metrics import mean_squared_error, r2_score
5 X = np.array([[1], [2], [3], [4], [5], [6], [7], [8], [9], [10]])
6 y = np.array([2, 5, 10, 17, 26, 37, 50, 65, 82, 101])
7 linear_model = LinearRegression()
8 linear_model.fit(X, y)
9 y_linear = linear_model.predict(X)
10 poly = PolynomialFeatures(degree=2)
11 X_poly = poly.fit_transform(X)
12 poly_model = LinearRegression()
13 poly_model.fit(X_poly, y)
14 y_poly = poly_model.predict(X_poly)
15 print("Linear Regression")
16 print("MSE:", mean_squared_error(y, y_linear))
17 print("R2 Score:", r2_score(y, y_linear))
18 print("\nPolynomial Regression")
19 print("MSE:", mean_squared_error(y, y_poly))
20 print("R2 Score:", r2_score(y, y_poly))
```

Run exp 9

```
/Users/moshika/PycharmProjects/PythonProject/.venv/bin/python /Users/moshika/PycharmProjects/PythonProject/exp 9.py
Linear Regression
MSE: 52.80000000000003
R2 Score: 0.9497645211930926

Polynomial Regression
MSE: 1.1525257825278982e-28
R2 Score: 1.0
```

Result: Linear and Polynomial Regression models were successfully implemented and compared based on their prediction performance.